

Employment

- Dec. 2022– **Senior Consultant, Energy Markets, Aurecon**, Sydney, Australia
- Consulting on energy markets, technoeconomics of green energy and electrification, and other commercial aspects of the energy transition. Working with the Decarbonisation team, I have:
- **Advised international and domestic clients on the fundamentals of entering and operating within the Australian energy market.** Areas I have worked in include grid connection, market services, requirements and obligations for market registration and trading, and contracting for electricity sale, supply, and hedging.
 - **Led development of Linear Programming models in Python (LP optimisation) to simulate optimal market trading of NEM energy assets**, estimating revenue potential and opportunity costs of capacity reservation for non-market services. Created constrained optimisation models for integrated systems such as standalone and hybrid BESS, V2G-enabled EV fleets, flexible hydrogen production and green chemical production precincts, electric smelters, gas turbines, CST+TES solar thermal, data centres, and other assets.
 - Built interactive financial models (Excel) to benchmark and compare energy assets and electrification candidate scenarios.
 - Developed NEM analyses drawing on data from the AEMO Market Management System (MMS) database using third-party tools (IES' NeoExpress, QueryTool), using Python and SQL.
 - Co-developed and maintained a dispatch trading model of the NEM (using IES' Prophet) based on the AEMO Integrated Systems Plan with original extensions to improve fidelity around hydro topology and add stochastic (Markov) climate variations.
- 2020–2022 **Postdoctoral Research Fellow, UNSW (School of Physics)**, Sydney, Australia
- Postdoctoral research on exciton spin physics and excitonic logic, relevant to next-generation solar panels and some forms of quantum computing. This work combined experimental measurements, mathematical modelling (analytical model-building and numerical simulations of classical and quantum dynamics in Python), and theoretical analysis of logic and computing.
- Independently managed a research theme within the larger group.
 - Experimental studies (spin resonance) of excitonic materials.
 - Built analytical models of excitonic processes using mathematical skills (calculus, linear algebra).
 - High performance computing on local and national computing clusters (NCI Gadi, UNSW Katana).
 - Published research in top journals on and presented work at local and international conferences.

Education

- 2016–2020 **PhD in Chemistry, UNSW**, Sydney, Australia
- Research on directing the movement of molecules in solution. Themes included control of molecules with light, molecular logic, emergence of complex systems from simple interactions. [Thesis available.](#)
- Granted **UNSW Dean's Award for an Outstanding PhD Thesis**, awarded to candidates that "...produce a thesis that requires only minimal corrections, receives outstanding and/or excellent levels of achievement for all examination criteria, and [...] is in the top 10% of PhD theses".
 - Research funded through award of competitive RTP Scholarship
 - Developed new experimental techniques and data analyses in physical chemistry.
 - Built mathematical models for chemical interactions, programmed numerical simulations (Python), and used experimental data to fit and understand intermolecular processes.
 - Published research in high-impact journals and presented at major conferences.
- 2018 **Research Exchange, University of Groningen**, Groningen, Netherlands
- Research on new molecular devices with Prof. Ben Feringa, 2016 Nobel Laureate in Chemistry.
- 2011–2014 **Bachelor of Science (Honours in Chemistry), University of Sydney**, Sydney, Australia
- First Class Honours. Other courses in Physics, Mathematics, Engineering, Linguistics.

References

References can be provided upon request.